

IDS Code

CJD

2025-11-27

Assignment brief • Perform exploratory data analysis to assess how 6 continents in the world (all continents except Antarctica) are faring with respect to the following specific targets set out by the UN:

1. Sustain per capita economic growth in accordance with national circumstances and, in particular, at least 7 per cent gross domestic product growth per annum in the least developed countries.
2. By 2020, substantially reduce the proportion of youth not in employment, education or training.

Find another CSV file to add to analysis

Loading libraries:

```
library(dplyr)

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(ggplot2)
```

We must set the wd and import the 3 CSV files as data frames. (note eet means education, employment, and training)

```
setwd("/Users/charliedicker/Downloads/data sets")

continents <- read.csv("continents-according-to-our-world-in-data.csv")
gdp_per_capita_df <- read.csv("gdp-per-capita-worldbank.csv")
youth_not_in_eet <- read.csv("youth-not-in-education-employment-training.csv")
ldc_df <- read.csv("LDCs.csv")
```

We now clean the data, getting rid of any irrelevant data

```

# The year is irrelevant in continents, as the only thing this df needs to be used for is assigning a c
continents$Year <- NULL
continents$Entity <- NULL

# We now assign the continent to each country in gdp per capita
gdp_per_capita_df <- gdp_per_capita_df %>%
  left_join(continents, by = "Code")

# Same thing for youth not in eet
youth_not_in_eet <- youth_not_in_eet %>%
  left_join(continents, by = "Code")

# The continents table is now redundant
rm(continents)

```

Still cleaning (renaming columns to nicer names)

```

# Rename a column in gdp per capita
gdp_per_capita_df <- gdp_per_capita_df %>%
  rename("Gdp_per_capita" = "GDP.per.capita..PPP..constant.2017.international...")

# Rename a column in youth not in eet
youth_not_in_eet <- youth_not_in_eet %>%
  rename("Perc_youth_not_in_eet" = "Share.of.youth.not.in.education..employment.or.training..total....o

```

Dropping values before the year 2014, and if continent is NA

```

# Drop all values before 2014, as these UN goals were set in 2015
gdp_per_capita_df <- gdp_per_capita_df %>%
  filter(Year >= 2014 & !is.na(Continent))

youth_not_in_eet <- youth_not_in_eet %>%
  filter(Year >= 2014 & !is.na(Continent))

```

Calculating Economic Growth and change in eet of each country

```

gdp_per_capita_df <- gdp_per_capita_df %>%
  mutate(Growth = ifelse(lag(Entity) != Entity, NA, (Gdp_per_capita - lag(Gdp_per_capita)) / lag(Gdp_per_

youth_not_in_eet <- youth_not_in_eet %>%
  mutate(Perc_change = ifelse(lag(Entity) != Entity, NA, (Perc_youth_not_in_eet - lag(Perc_youth_not_in

```

This chunk is now to add a flag to the gdp per capita dataframe, labelling if a country is an LDC or not

```

gdp_per_capita_df <- gdp_per_capita_df %>%
  mutate(Ldc = Entity %in% ldc_df$Entity)

```

REF: 25/11/2025 asked ChatGPT “what would the definition of”least developed countries” mean in terms of gdp per capita (PPP)”

26/11/2025 asked ChatGPT “create me a downloadable csv file, containing all ldcs recognised by the un between the years of 2015 and 2021, with the title of the column being”Entity” ”